413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 1

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1 Probability, Uncertainty, and Counting

1.1 Why Probability and Statistics?

Many mathematical models describe situations in which the outcome is known once the inputs are specified. These are deterministic functions, whose outcome is fixed given an input. However, in many real-world problems, we must reason in the presence of **uncertainty** arising from measurement errors, insufficient samples, and intrinsic randomness of the underlying system. Probability and statistics provide a mathematical framework for reasoning about **uncertainty** and learning from **data**. Although closely related, they address two complementary questions.

In **probability**, we assume that a probabilistic model is known and ask what kinds of observations it is likely to generate:

$$\boxed{\text{Model} \quad \longrightarrow \quad \text{Data}}$$

On the other hand, in **statistics**, we observe data and try to infer properties of the unknown process or distribution that generated it:

$$\boxed{\text{Data} \quad \longrightarrow \quad \text{Model}}$$

Given the ubiquitous nature of randomness and uncertainty associated with most worldly events, **probability** and **statistics** are used for many applications. A non-exhaustive list of application domains include genetics and other biological sciences, physics, fluid dynamics, quantum mechanics, finance and economics, political and social sciences, and many more.

A central challenge is that our intuition about uncertainty is often very unreliable, as most results in statistics are very counter-intuitive. It contains many examples where an intuitively plausible answer is incorrect. We therefore want a precise mathematical framework for questions of the form

> *What are all the possible outcomes, and how likely is a particular collection of outcomes ?*

This leads naturally to the concepts of **sample spaces**, **events**, and **probabilities**.

1.2 Sample Spaces and Events

An **experiment** is any procedure whose outcome is not known beforehand. The word "experiment" is interpreted broadly: it could mean tossing a coin, rolling a dice, measuring a quantity of interest in a lab, observing a financial return, or recording the outcome of any uncertain process.

The **sample space**, denoted by Ω, is the set of all possible outcomes of the experiment:

$$\Omega = \{\text{all possible outcomes}\}.$$

An **event** A is a subset of the sample space,

$$A \subseteq \Omega.$$

For example, if we toss a coin twice,

$$\Omega = \{HH, HT, TH, TT\}.$$

The event that both tosses are tails is

$$A = \{TT\}.$$

1.3 A Lazy Definition of Probability

Suppose that the sample space Ω is **finite** and that all outcomes are **equally likely**. Then the probability of an event A is

$$\boxed{P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes} \in \text{A}}{\text{total number of outcomes}}.}$$

For example, for two fair coin tosses,

$$P(A = TT) = \frac{1}{4}.$$

IMPORTANT: There are two assumptions underlying the naive definition of probability:

1. The sample space is **finite**.
2. All outcomes in the sample space are **equally likely**.

Thus, whenever we use the lazy definition, we must first justify why the outcomes can be regarded as equally likely. Once this assumption is justified, estimating probabilities of events becomes a **counting problem**. The next question is therefore:

How do we efficiently count the number of possible outcomes?

1.4 Counting rule

Suppose an experiment consists of r successive choices. There are n_1 possible outcomes for the first choice, and for *each* such outcome there are n_2 possible outcomes for the second choice, and so on. If there are n_r possible outcomes for the rth choice, then the total number of possible outcomes is

$$\boxed{n_1 n_2 \cdots n_r.}$$

A useful way to visualize the multiplication rule is through a **tree diagram**: each successive choice creates another collection of branches.

Example. Suppose a furniture shop offer chairs with six different colors and three different materials. The number of possible color-material combination is

$$6 \times 3 = 18.$$

The number of possibilities can grow very quickly. For example, if we make 10 successive binary choices, then the number of possible outcomes are 2^{10}.

Thus, even simple experiments can have sample spaces that are far too large to enumerate explicitly. Counting rules provide compact mathematical expressions that allow us to determine their size without listing every outcome.

2 General Principles for Counting

Counting problems can be deceptively subtle. Before doing any algebra, it is useful to keep a few principles in mind. Many probability calculations begin with a simple question:

$$\boxed{\text{How many different outcomes are possible?}}$$

Suppose we choose k objects from n distinct objects, **without replacement**, and the **order does not matter**. The number of possible selections is the **binomial coefficient**

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}.}$$

For example, the number of possible five-card poker hands is

$$\boxed{\binom{52}{5}.}$$

1. Check simple and extreme cases. Suppose we derive a formula depending on parameters such as n and k. Before trusting it, check cases such as

$$k = 0, \qquad k = 1, \qquad n = 1,$$

or the simplest nontrivial example. For example,

$$\binom{n}{0} = 1,$$

because there is exactly one way to choose no objects from n objects. *Such trivial cases are often enough to reveal an incorrect counting argument.*

2. Label objects when appropriate. If a problem involves n people or objects, it is often useful to imagine that they are labeled

$$1, 2, \ldots, n.$$

Even if two physical objects look identical, they may still correspond to distinct outcomes of an experiment.

3. Be careful about overcounting. For example, suppose 10 people are divided into a team of 4 and a team of 6. Choosing the team of 4 completely determines the team of 6, so

$$\binom{10}{4} = \binom{10}{6}$$

is the number of possible divisions. In general,

$$\boxed{\binom{n}{k} = \binom{n}{n-k}.}$$

To choose k people from a group of n is equivalent to choosing the $n-k$ people who are *not* selected. Both sides therefore count the same thing.

In contrast, suppose 10 people are divided into *two unlabeled teams* of 5. Then $\binom{10}{5}$ counts every division twice: choosing the first group of five or choosing its complement gives the same pair of teams. Hence, $\frac{1}{2}\binom{10}{5}$ is the correct count. Thus, "does order matter?" is often really a question about whether two descriptions correspond to genuinely different outcomes.

4: Choosing a committee and its president. Suppose we want to choose a committee of k people from n people and designate one committee member as president. One approach is:

$$\underbrace{\binom{n}{k}}_{\text{choose committee}} \quad \underbrace{k}_{\text{choose president}}.$$

Alternatively, first choose the president and then choose the remaining $k-1$ committee members:

$$\underbrace{n}_{\text{choose president}} \quad \underbrace{\binom{n-1}{k-1}}_{\text{choose remaining members}}.$$

Since both procedures count the same outcomes,

$$\boxed{k\binom{n}{k} = n\binom{n-1}{k-1}.}$$

5: Vandermonde's Identity. Suppose there are two groups containing m and n people, respectively, and we want to select k people in total. Directly, there are

$$\binom{m+n}{k}$$

ways. Alternatively, suppose exactly j selected people come from the first group. Then $k-j$ must come from the second group. For a fixed j, there are

$$\binom{m}{j}\binom{n}{k-j}$$

possibilities. Summing over all possible j gives

$$\boxed{\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}.}$$

This is **Vandermonde's identity**. The identity follows immediately by counting the same selection problem in two different ways.

Connection to Pascal's Triangle

The same numbers appear as coefficients in the **binomial theorem**:

$$\boxed{(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.}$$

For example,

$$(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.$$

The coefficients can be read directly from **Pascal's triangle**:

$$\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \quad \longleftarrow \ \tbinom{4}{0}, \tbinom{4}{1}, \dots, \tbinom{4}{4} \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}$$

Each entry is obtained by adding the two entries above it:

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.}$$

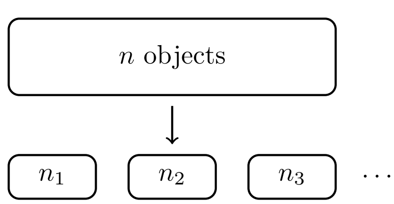
From Two Groups to Many Groups

The binomial coefficient divides n objects into **two groups**:

$$\underbrace{k}_{\text{selected}} \quad + \quad \underbrace{n-k}_{\text{remaining}}.$$

What if instead we divide the objects into m groups of sizes

$$n_1, n_2, \dots, n_m, \qquad n_1 + \cdots + n_m = n?$$



The number of possible divisions is the **multinomial coefficient**

$$\boxed{\binom{n}{n_1, n_2, \ldots, n_m} = \frac{n!}{n_1! n_2! \cdots n_m!}}.$$

Why? Choose each group sequentially:

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots.$$

Writing out the factorials produces cancellations:

$$\frac{n!}{n_1!(n-n_1)!}\frac{(n-n_1)!}{n_2!(n-n_1-n_2)!}\cdots = \boxed{\frac{n!}{n_1!\cdots n_m!}}.$$

The central counting principle is

Choose k objects from n		Divide n objects into several groups
$\binom{n}{k}$	$\longrightarrow$	$\binom{n}{n_1, \ldots, n_m}$.

Binomial and multinomial coefficients count possibilities; when combined with probabilities for the individual outcomes, they lead to the binomial and multinomial probability distributions.

2.1 Beyond the Naive Definition of Probability

The naive definition

$$P(A) = \frac{|A|}{|\Omega|}$$

requires a finite sample space whose outcomes are equally likely. We now want a definition that also allows

- outcomes that are not equally likely, and
- sample spaces with infinitely many possible outcomes.

As discussed before, the **probability space** consists of a sample space Ω together with a probability function P that assigns to each event $A \subseteq \Omega$ a number

$$P(A) \in [0, 1].$$

Thus, we may think of

$$P : \{\text{events in } \Omega\} \longrightarrow [0, 1].$$

2.2 Axioms of Probability

The probability function $P(A)$ satisfies the following fundamental rules.

Axiom : Impossible and certain events.

$$\boxed{P(\emptyset) = 0, \qquad P(\Omega) = 1.}$$

The empty event is impossible, whereas some outcome in the sample space must occur.

Axiom 2: Additivity for disjoint events. If

$$A_1, A_2, A_3, \ldots$$

are pairwise disjoint events, then

$$\boxed{P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).}$$

For two disjoint events A and B, this reduces to

$$P(A \cup B) = P(A) + P(B), \qquad A \cap B = \emptyset.$$